4-42. The life in hours of a thermocouple used in a furnace is known to be approximately normally distributed, with standard deviation $\sigma = 20$ hours. A random sample of 15 thermocouples resulted in the following data: 553, 552, 567, 579, 550, 541, 537, 553, 552, 546, 538, 553, 581, 539, 529.

(a) Is there evidence to support the claim that mean life exceeds 540 hours? Use a fixed-level test with $\alpha = 0.05$.

(b) What is the P-value for this test?

(c) What is the β-value for this test if the true mean life is 560 hours?

(d) What sample size would be required to ensure that β does not exceed 0.10 if the true mean life is 560 hours?

(e) Construct a 95% one-sided lower CI on the mean life.

(f) Use the CI found in part (e) to test the hypothesis.

4-44. Suppose that in Exercise 4-42 we wanted to be 95% confident that the error in estimating the mean life is less than 5 hours. What sample size should we use?

4-54. An article in the *ASCE Journal of Energy Engineering* (Vol. 125, 1999, pp. 59–75) describes a study of the thermal inertia properties of autoclaved aerated concrete used as a building material. Five samples of the material were tested in a structure, and the average interior temperature (°C) reported was as follows: 23.01, 22.22, 22.04, 22.62, and 22.59.

(a) Test the hypotheses H_0: $\mu = 22.5$ versus H_1: $\mu \neq 22.5$, using $\alpha = 0.05$. Use the P-value approach.

(b) Check the assumption that interior temperature is normally distributed.

(c) Find a 95% CI on the mean interior temperature.

(d) What sample size would be required to detect a true mean interior temperature as high as 22.75 if we wanted the power of the test to be ~~a~~ at least 0.9? Use the sample standard deviation s as an estimate of σ.

4-59. In building electronic circuitry, the breakdown voltage of diodes is an important quality characteristic. The breakdown voltage of 12 diodes was recorded as follows: 9.099, 9.174, 9.327, 9.377, 8.471, 9.575, 9.514, 8.928, 8.800, 8.920, 9.913, and 8.306.

(a) Check the normality assumption for the data.

(b) Test the claim that the mean breakdown voltage is less than 9 volts with a significance level of 0.05.

(c) Construct a 95% one-sided upper confidence bound on the mean breakdown voltage.

(d) Use the bound found in part (c) to test the hypothesis.

(e) Suppose that the true breakdown voltage is 8.8 volts; it is important to detect this with a probability of at least 0.95. Using the sample standard deviation to estimate the population standard deviation and a significance level of 0.05, determine the necessary sample size.

4-65. Cloud seeding has been studied for many decades as a weather modification procedure (for an interesting study of this subject, see the article in *Technometrics*, "A Bayesian Analysis of a Multiplicative Treatment Effect in Weather Modification," Vol. 17, 1975, pp. 161–166). The rainfall in acre-feet from 20 clouds that were selected at random and seeded with silver nitrate follows: 18.0, 30.7, 19.8, 27.1, 22.3, 18.8, 31.8, 23.4, 21.2, 27.9, 31.9, 27.1, 25.0, 24.7, 26.9, 21.8, 29.2, 34.8, 26.7, and 31.6.

(a) Can you support a claim that mean rainfall from seeded clouds exceeds 25 acre-feet? Use $\alpha = 0.01$. Find the P-value.

(b) Check that rainfall is normally distributed.

(c) Compute the power of the test if the true mean rainfall is 27 acre-feet.

(d) What sample size would be required to detect a true mean rainfall of 27.5 acre-feet if we wanted the power of the test to be at least 0.9?

(e) Explain how the question in part (a) could be answered by constructing a one-sided confidence bound on the mean ~~diameter~~.

4-71. The percentage of titanium in an alloy used in aerospace castings is measured in 51 randomly selected parts. The sample standard deviation is $s = 0.37$.

(a) Test the hypothesis $H_0: \sigma = 0.35$ versus $H_1: \sigma \neq 0.35$ using $\alpha = 0.05$. State any necessary assumptions about the underlying distribution of the data.

(b) Find the P-value for this test.

(c) Construct a 95% two-sided CI for σ.

(d) Use the CI in part (c) to test the hypothesis.

4-75. Large passenger vans are thought to have a high propensity of rollover accidents when fully loaded. Thirty accidents of these vans were examined, and 11 vans had rolled over.

(a) Test the claim that the proportion of rollovers exceeds 0.25 with $\alpha = 0.10$.

(b) Suppose that the true $p = 0.35$ and $\alpha = 0.10$. What is the β-error for this test?

(c) Suppose that the true $p = 0.35$ and $\alpha = 0.10$. How large a sample would be required if we want $\beta = 0.10$?

(d) Find a 90% traditional lower confidence bound on the rollover rate of these vans.

(e) Use the confidence bound found in part (d) to test the hypothesis.

(f) How large a sample would be required to be at least 95% confident that the error on p is less than 0.02? Use an initial estimate of p from this problem.

4-76. A random sample of 50 suspension helmets used by motorcycle riders and automobile race-car drivers was subjected to an impact test, and on 18 of these helmets some damage was observed.

(a) Test the hypotheses $H_0: p = 0.3$ versus $H_1: p \neq 0.3$ with $\alpha = 0.05$.

(b) Find the P-value for this test.

(c) Find a 95% two-sided traditional CI on the true proportion of helmets of this type that would show damage from this test. Explain how this confidence interval can be used to test the hypothesis in part (a).

(d) Using the point estimate of p obtained from the preliminary sample of 50 helmets, how many helmets must be tested to be 95% confident that the error in estimating the true value of p is less than 0.02?

(e) How large must the sample be if we wish to be at least 95% confident that the error in estimating p is less than 0.02, regardless of the true value of p?

4-89. Consider the helmet data given in Exercise 4-76. Calculate the 95% Agresti-Coull two-sided CI from equation 4-76 and compare it to the traditional CI in the original exercise.

4-95. Consider the breakdown voltage of diodes described in Exercise 4-59.

(a) Construct a 99% PI for the breakdown voltage of a single diode.

(b) Find a tolerance interval for the breakdown voltage that includes 99% of the diodes with 99% confidence.